



PROBLEM II [HOSPITAL INVENTORY SYSTEM]

DS331: Systems Modeling and Simulation

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Table of contents

Report Components

- Problem formulation & Objectives.....
- System Components.....
- System analysis including cumulative distribution tables simulation table (for 20 days).....
- Experimental Design Parameters.....
- Justification of experiment parameters values.....
- Results Analysis: Using graphs & discussions stating the results
- Conclusion.....

Simulation Program

- Coding Style (naming convention, comments)
- GUI and Data Visualization (graphs).....
- Correct computation and results for:
- The average ending units in first floor inventory and the basement inventory.....
- The number of days when a shortage condition occurs.....
- Does the theoretical average demand of boxes match the experimental one?.....
- Does the theoretical average lead time of the lead time distribution match the experimental one?.....
- Is there a better value for the review period variable N to minimize the shortages of medical supplies boxes?.....
- Is there better value for the maximum capacity M of the basement inventory to minimize the shortages of medical supplies boxes?.....
- Extra features in the simulator (for example: generic or extra statistics)

Problem formulation & Objectives

Problem formulation:

1.1 Understanding the Problem

- The hospital operates a two-tier inventory system to manage medical supplies efficiently:
 1. First Floor Inventory: Directly provide patient rooms, with each room consuming one box of supplies daily. It has a maximum capacity of 10 boxes. When it runs out of supplies, shortages occur.
 2. Basement Inventory: Acts as the main storage, holding up to 30 boxes. It replenishes the first-floor inventory on request. Shortages in the basement occur when it cannot fulfill the first-floor requests due to low stock.
- To replenish the basement inventory, the hospital places orders periodically based on the Review Period (N). Restocking orders are subject to random Lead Times, which affect when the supplies are received. The demand for supplies depends on the number of rooms occupied, which varies daily according to probability distribution.

1.2 Reformulation of the Problem

The hospital must optimize its inventory system to balance supply and demand while minimizing shortages. This involves:

- Managing daily variability in demand (room occupancy).
- Dealing with lead time uncertainty during restocking.
- Determining the ideal review period N and basement capacity M to minimize shortages while ensuring system efficiency.

A simulation-based approach will model the interactions between the first floor and basement inventories and assess the impact of these uncertainties on inventory performance.

2. Objectives and Overall Project Plan

2.1 Main Objective

To design and evaluate an inventory management system that minimizes shortages while maintaining efficient operations.

2.2 Specific Objectives

1. Calculate the average ending inventory levels for the first floor and basement inventories.
2. Determine the number of days when shortages occur.
3. Compare simulated demand with the theoretical demand distribution based on room occupancy.
4. Compare simulated lead times with the theoretical lead time distribution.
5. Identify the optimal review period N to minimize shortages.
6. Evaluate the maximum basement capacity M required to reduce shortages effectively.

Is Simulation Appropriate?

Simulation is the most suitable method because:

- Complexity: The problem involves random daily demand and lead times, which are challenging to address using static calculations.
- Scenarios Testing: Simulation enables testing various review periods N and basement capacities M without real-world risks.
- Cost Effectiveness: It avoids the operational risks and costs of testing changes directly in the hospital system.

Costs and Benefits

Costs:

- Direct Costs: Developing the simulation model and computing resources for running the model.
- Indirect Costs: Time spent by staff in data analysis and decision-making.

Benefits:

- Avoids operational disruptions and shortages in the real system.
- Provides insights to optimize inventory policies and ensure continuous supply for patients.
- Balances storage costs with shortage risks through informed decision-making.

Questions to Be Answered After Finish The Simulation:

1. What are the average ending inventory levels for the first floor and basement inventories?
2. How often do shortages occur, and what are the causes?
3. Does the simulated demand match the theoretical demand distribution?
4. Does the simulated lead time align with the theoretical distribution?
5. What is the optimal review period N for minimizing shortages?
6. What is the best maximum capacity M for the basement inventory to minimize shortages?

System Components

Entities	First Floor Inventory, Basement Inventory, Order System, Patient Rooms.
Attributes	Current stock, Capacity, Order Quantity, Lead Time, Occupancy.
Activities	Daily demand calculation, replenishing inventory, restocking basement, receiving orders.
State Variables	Current stock (for both floors), Order status, Shortage indicator
Events	Daily demand event, low stock trigger event, order placement event, order fulfillment event, shortage event.

System analysis

No. of rooms occupied	Probability	Cumulative probability	Random -digit assignment
1	0.1	0.10	0-10
2	0.15	0.25	11-25
3	0.35	0.60	26-60
4	0.2	0.80	61-80
5	0.2	1.00	81-00

"Table 1"

Lead Time	Probability	Cumulative probability	Random -digit assignment
1	0.4	0.40	0-40
2	0.35	0.75	41-75
3	0.25	1.00	76-00

"Table 2"

Simulation Table

Day	Beginning inventory first floor	Beginning basement inventory	Random digit first floor demand	First floor demand	Ending inventory First floor	Ending inventory Basement	Shortage First floor inventory	Order quantity for first floor	Order quantity For basement inventory	Random digit for lead time	Lead time
1	4	30	64	4	0	30	0	10	0	-	0
2	10	20	8	1	9	20	0	0	0	-	0
3	9	20	48	3	6	20	0	0	0	-	0
4	6	20	62	4	2	20	0	0	0	-	0
5	2	20	97	5	0	20	3	10	0	-	0
6	10	10	22	2	5	10	0	0	20	40	1
7	5	10	19	2	3	10	0	0	0	-	0
8	3	30	18	2	1	30	0	0	0	-	0
9	1	30	35	3	0	30	2	10	0	-	0
10	10	20	78	4	4	20	0	0	0	-	0
11	4	20	52	3	1	20	0	0	0	-	0
12	1	20	58	3	0	20	2	10	20	63	2
13	10	10	83	5	3	10	0	0	0	-	1
14	3	10	98	5	0	10	2	10	0	-	0
15	10	20	60	3	5	20	0	0	0	-	0
16	5	20	11	2	3	20	0	0	0	-	0
17	3	20	29	3	0	20	0	10	0	-	0
18	10	10	53	3	7	10	0	0	20	77	3
19	7	10	53	3	4	10	0	0	0	-	2
20	4	10	80	4	0	10	0	10	0	-	1

Experimental Design Parameters

- Review Period N
- Maximum Basement Capacity M
- Lead Time Distribution
- Daily Demand Distribution
- Initial Inventory Levels

Why This Parameters Specially?

1. Review Period N

The review period N determines how often inventory levels are checked and replenished. It balances the need for frequent updates to avoid stockouts with the cost of placing orders. A well-chosen review period helps optimize inventory flow, ensuring adequate stock without incurring unnecessary costs.

2. Maximum Basement Capacity M

The maximum basement capacity M sets the upper limit for the stock in the basement inventory. It ensures the system has enough backup stock to meet demand while preventing overstocking, which could lead to excess carrying costs and storage inefficiencies. This capacity is critical for maintaining a buffer to prevent shortages.

3. Lead Time Distribution

The lead time distribution models the variability in the time it takes to receive orders. Since lead times are unpredictable, this distribution accounts for potential delays in the supply chain. It helps simulate realistic conditions, enabling the system to plan for and manage replenishment delays effectively.

4. Daily Demand Distribution

The daily demand distribution reflects the variation in demand based on the number of rooms occupied. It accounts for fluctuations in consumption, ensuring the system can dynamically respond to varying daily needs. Understanding this helps in maintaining adequate stock levels and avoiding shortages.

Results Analysis

In case of 20 Day

1. The Average of First Floor Ending
= The ending units in inventory /number of days= 2.33
2. The Average of Basement Floor Ending
= The ending units in inventory /number of days = 18.1
3. The number of days when a shortage condition occurs= 5 Days
- 4- Distribution of demand:

The theoretical average for demand = 3.25

The experimental average for demand = 3.05

The theoretical average demand of boxes does not match the experimental one.

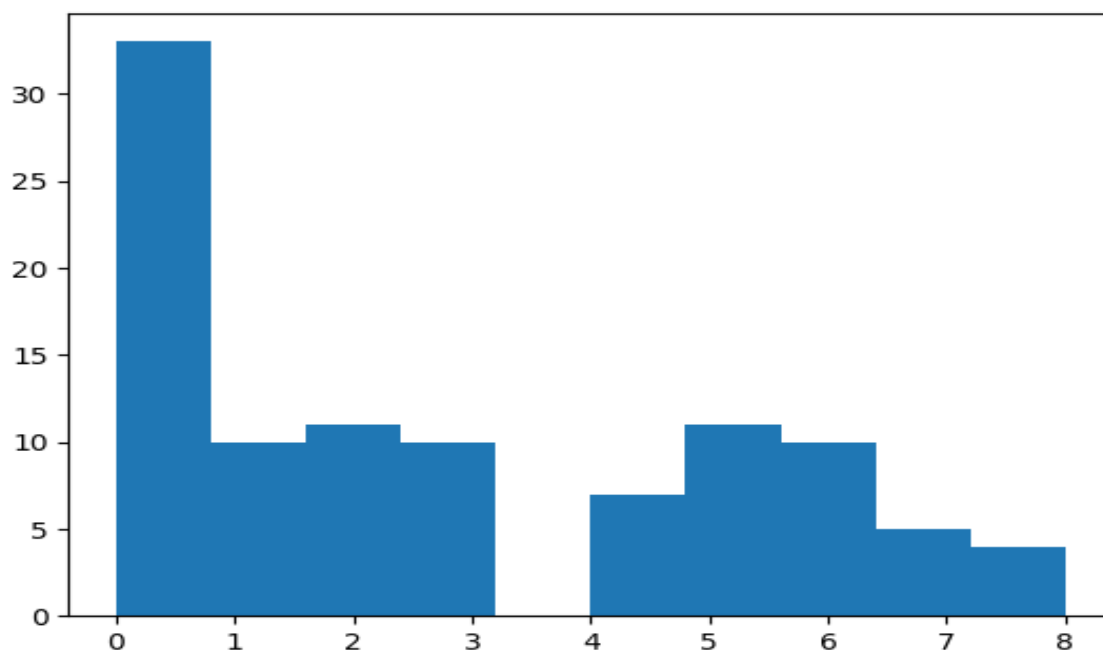
5- Lead Time Distribution:

The theoretical average for Lead Time = 1.85

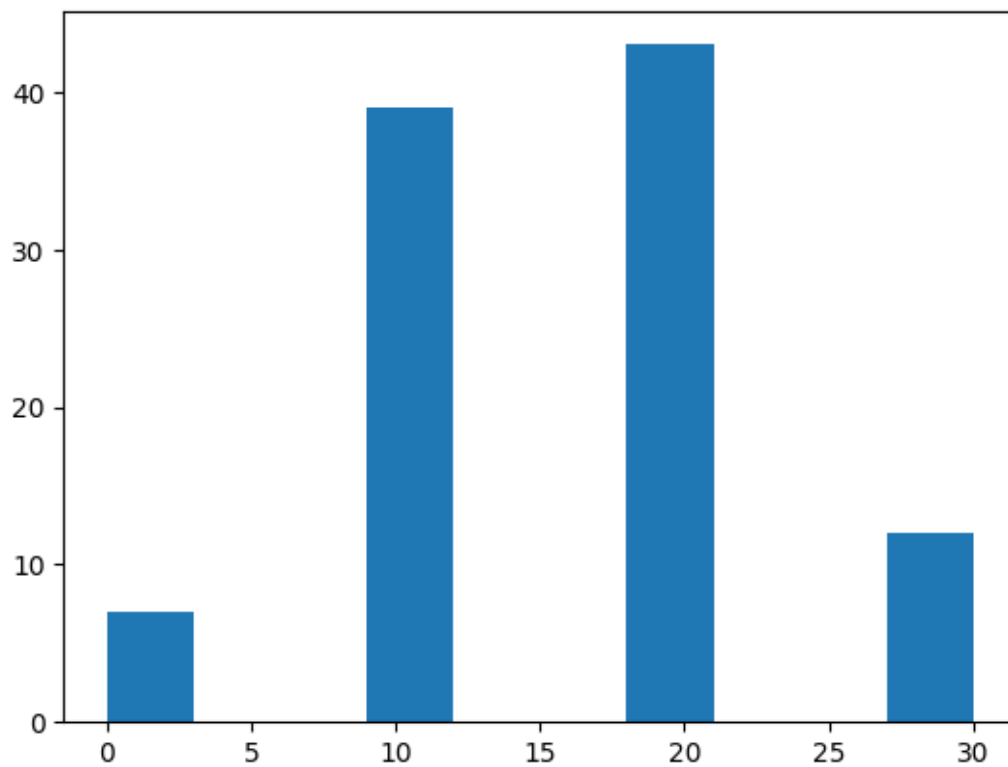
The experimental average for Lead Time = 1.67

Now let's run for 100 days

1. The Average of First Floor Ending =2.69



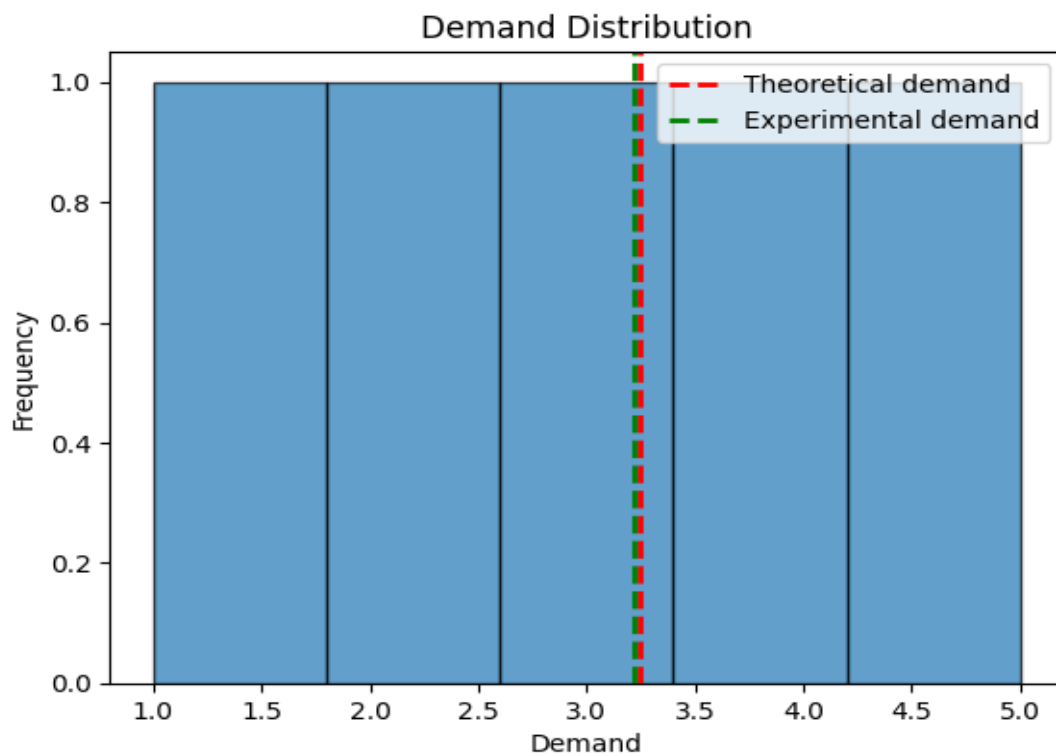
2. The Average of Basement Floor Ending = 15.94



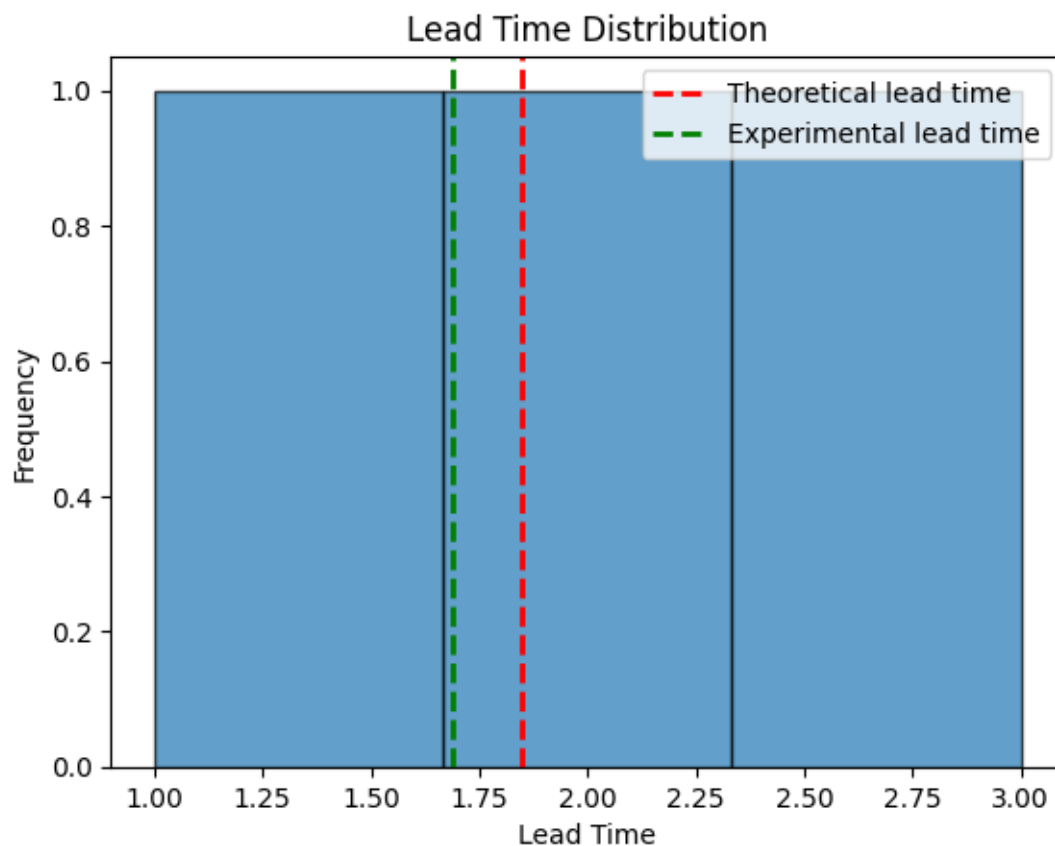
3. The number of days when a shortage condition occurs= 22 Days

4. The theoretical average for demand= 3.25

5. The experimental average for demand = 3.22



6. The theoretical average for Lead Time =1.85
7. The experimental average for Lead Time = 1.69



From those results we can get averages of Average by run 30 trails according to 100 day:

The Average of First Floor Ending= 2.687

The Average of Basement Floor Ending= 16.595

The experimental average for demand= 3.1753

The experimental average for Lead Time= 1.846

Result analysis Discussion And Conclusion:

❖ Comparison Between Theoretical and Experimental Average Demand

- Results:
 - The theoretical average demand = 3.25 boxes/day.
 - The experimental average demand = 3.1753 boxes/day.

From that we note that the experimental demand (3.1753) is very close to the theoretical demand (3.25), showing that the simulation accurately reflects the expected patient room occupancy and daily box consumption.

❖ Comparison Between Theoretical and Experimental Average Lead Time

- Results:
 - The theoretical average lead time = 1.85 days.
 - The experimental average lead time = 1.846 days.

From that we note that the experimental lead time is almost identical to the theoretical lead time. This indicates that the simulation correctly captures the random variation in lead times as per the distribution provided.

❖ What about the Optimal Value for the M and the N?

Shortages and Inventory Policy The increasing number of shortage days in longer simulations points to the fact that the current parameters for the review period N and maximum capacity M might not be optimal.

- Increasing M could help reduce shortages by providing a larger inventory buffer, but it would raise the carrying costs.
- Decreasing N could reduce shortages by allowing more frequent inventory replenishments, but it would increase ordering costs.

This trade-off between holding costs and ordering costs needs further investigation to find the optimal values for N and M that minimize shortages while controlling costs effectively.

In conclusion, the simulation provides valuable insights into improving inventory management by identifying factors contributing to shortages and inefficiencies. Adjusting parameters like review periods or storage capacity can help reduce shortages and enhance overall system performance. The close alignment between theoretical and experimental results confirms the model's reliability, offering a strong basis for optimizing inventory policies and ensuring better resource availability.

Our code on Google collab:

<https://colab.research.google.com/drive/1xz7Xyclwr36SdnrWEI53RSiD4KIKGyb2?usp=sharing>